From Transaction Cost to Compounding Architecture: A Theory of the Firm in an AI Economy

Abstract

Classical theories of the firm — from Coase's transaction cost framework to Williamson's governance logic and Porter's structural positioning — explain organisational hierarchy as a response to market friction. This paper argues that artificial intelligence systematically erodes the frictions on which these explanations rest: AI reduces search costs, compresses information asymmetries, automates coordination, and enables real-time optimisation across economic subsystems. If friction was the original justification for hierarchy, its erosion demands a new account of why firms persist and what organisational form generates advantage.

We propose that in AI-mediated markets, firms persist not primarily to minimise transaction cost, but to govern and rebalance compounding economic architectures under constraint. Extending the Customer Economic Architecture (Box) Model, we formalise firm value as a multiplicative system across four interdependent domains: economic segment precision (WHO), contribution margin architecture (WHAT), cost physics and operational leverage (HOW), and customer lifetime value compounding (CLV). In this formulation, local AI optimisation of each domain is increasingly tractable; but cross-domain constraint arbitration — the identification of the binding dimension and the deliberate reallocation of capital toward it, at the expense of locally efficient alternatives —

remains irreducibly a governance problem that market mechanisms and algorithmic systems cannot resolve.

We advance five formal propositions defining the AI-era firm as a constraint-orchestrating governance structure, and identify the conditions under which hierarchy produces advantages that AI-augmented markets cannot replicate. The theory generates a research agenda at the intersection of strategic management, organisational theory, and digital economics, and has direct implications for organisational design, capital allocation governance, and the appropriate scope of algorithmic decision-making.

Keywords: *theory of the firm; transaction cost economics; artificial intelligence; compounding architecture; Theory of Constraints; customer lifetime value; multiplicative systems; dynamic capabilities; governance*

JEL Classification: L22, D23, L10, M21, O33

1. INTRODUCTION

The dominant theories of the firm share a common architecture: they explain organisational hierarchy as a solution to a problem that markets cannot efficiently resolve. Coase established that firms arise when the transaction costs of market exchange exceed those of internal coordination. Williamson extended this logic to incomplete contracts and the hazards of opportunistic behaviour under asset specificity. Jensen and Meckling formalised the agency costs that arise when

ownership and control separate. Porter reframed the firm as a structural position within a system of industry forces that determine long-run profit potential.

Across these traditions, a common premise holds: firms exist because markets are imperfect. The friction in market exchange — costly search, asymmetric information, incomplete contracting, opportunism — creates the economic space within which hierarchy produces a superior governance outcome.[1]

Artificial intelligence systematically challenges this premise. AI reduces search costs toward near-zero in many domains. Predictive analytics compresses information asymmetry between transacting parties. Automated contracting and smart execution reduce incomplete contract risk. Algorithmic matching achieves coordination at scales and speeds that manual hierarchy cannot approach. If friction was the original justification for the firm, its progressive erosion by AI demands a new account of what organisational form generates advantage and why.

This challenge is not merely theoretical. The empirical evidence of AI's economic impact — documented by Brynjolfsson and McAfee on productivity, by Agrawal, Gans and Goldfarb on prediction cost, and by Goldfarb and Tucker on digital market structure — suggests that the conditions assumed by classical firm theory are increasingly non-representative of markets in which AI is a significant productive input. Recent large language model capabilities extend this further: AI systems can now perform not only routine coordination tasks but a substantial range of cognitive tasks previously considered firm-specific advantages.[2]

The question this paper addresses is therefore both positive and normative: if AI erodes the frictions that classical theory used to justify hierarchy, what is the remaining function of the firm?

[1]Coase, R.H. (1937). The nature of the firm. Economica, 4(16), 386–405.

[2]Williamson, O.E. (1979). Transaction-cost economics: The governance of contractual relations. Journal of Law and Economics, 22(2), 233–261.

And what organisational architecture generates durable advantage in an environment where local optimisation of any individual competitive dimension becomes computationally tractable?

We propose a new account. In AI-mediated markets, firms persist not primarily to minimise transaction cost but to govern and rebalance compounding economic architectures under constraint. The central insight is that AI excels at local optimisation within defined objective functions. It does not, however, resolve the meta-problem of determining which objective should dominate at the system level — which dimension of the firm's economic architecture constitutes the binding constraint on total value creation, and which dimensions should be deliberately sacrificed in the short term to protect long-term compounding. This constraint arbitration function is irreducibly a governance problem.

The paper makes three primary contributions. First, it identifies the erosion of market friction by AI as a boundary condition that weakens the explanatory foundations of classical firm theory, and proposes a new account of firm persistence grounded in constraint governance rather than coordination substitution. Second, it formalises firm value as a multiplicative system across four interdependent domains, generating non-trivial predictions about the returns to architectural balance versus local optimisation. Third, it derives five formal propositions characterising the AI-era firm as a constraint-orchestrating governance structure and identifies the conditions under which this function generates advantages that market mechanisms cannot replicate.

The paper proceeds as follows. Section 2 analyses the erosion of classical firm theory under AI conditions. Section 3 introduces the multiplicative architecture framework. Section 4 analyses the specific ways in which AI enables and endangers each dimension. Section 5 presents five formal propositions. Section 6 develops the constraint governance model. Section 7 examines

boundary conditions and scope. Section 8 derives organisational design implications. Section 9 presents a formal mathematical appendix. Section 10 concludes with a research agenda.

2. THE EROSION OF CLASSICAL FOUNDATIONS UNDER AI

2.1 Transaction Cost Economics and the AI Boundary Condition

Coase's original formulation identified four cost categories that determine the boundary between firm and market: search costs (the cost of identifying potential trading partners and relevant prices), contracting costs (the cost of negotiating and writing agreements), monitoring costs (the cost of verifying compliance), and enforcement costs (the cost of resolving disputes). The firm internalises transactions when these costs, in aggregate, exceed the costs of internal hierarchy — managerial salaries, organisational overhead, and the agency costs of delegated authority.[3]

Williamson's refinement added asset specificity — the degree to which investments are non-redeployable outside a specific relationship — and opportunism as the behavioural microfoundation. When asset specificity is high and opportunism is possible, the hold-up problem creates bilateral dependency that market contracts cannot efficiently resolve, motivating internalisation. This framework generates a clear prediction: as asset specificity increases, vertical integration is predicted to increase.[4]

AI systematically reduces the cost categories on which these predictions rest. Algorithmic matching platforms reduce search costs toward zero for a growing range of standardised inputs. Predictive analytics reduces information asymmetry between transacting parties. Smart contract

[3]Jensen, M.C. & Meckling, W.H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. Journal of Financial Economics, 3(4), 305–360.
[4]Porter, M.E. (1980). Competitive Strategy. Free Press, New York.

architectures — whether blockchain-based or AI-monitored — reduce incomplete contract risk by automating contingent performance monitoring. In labour markets, platforms like Upwork and Fiverr, augmented by AI matching, demonstrate near-zero transaction costs for a range of cognitive task categories previously internalised within firms. Agrawal, Gans and Goldfarb formalise this as a reduction in prediction cost: AI makes the contingent events that contracts must specify increasingly foreseeable, reducing the incompleteness that motivates Williamsonian internalisation.[5]

The theoretical implication is direct: if AI reduces the four Coasian cost categories to near-zero in a domain, the transaction cost rationale for internalising that domain collapses. The firm's boundary should contract toward the residual — those activities where asset specificity, relational capital, and institutional commitment generate returns that AI-augmented markets cannot deliver. This is a boundary condition, not a refutation: classical firm theory remains valid within the domain it was designed to explain, but AI progressively narrows that domain.

2.2 Agency Theory and the Measurement Problem Under AI

Jensen and Meckling's agency framework identifies the costs arising from the separation of ownership and control: monitoring expenditures by principals, bonding costs incurred by agents, and the residual loss from imperfect alignment. The central problem is measurement: principals cannot perfectly observe agent effort and output, creating moral hazard. Managerial hierarchy partially resolves this through direct supervision, accounting systems, and incentive contract design.[6]

[5]Teece, D.J., Pisano, G. & Shuen, A. (1997). Dynamic capabilities and strategic management. Strategic Management Journal, 18(7), 509–533.

[6]Teece, D.J. (2007). Explicating dynamic capabilities: The nature and microfoundations of sustainable enterprise performance. Strategic Management Journal, 28(13), 1319–1350.

AI dramatically improves measurement capability. Real-time performance monitoring, predictive analytics for output forecasting, and AI-assisted audit systems reduce the information asymmetry between principal and agent. Acemoglu and Restrepo document how automation of monitoring tasks has already altered wage structures in knowledge work. If measurement improves sufficiently, the residual loss component of agency cost declines, reducing the efficiency advantage of hierarchical monitoring over arm's-length contracting.[7]

However, AI introduces a new agency problem that the original framework does not capture: the objective function misalignment between AI optimisation systems and the long-run interests of the firm. An AI system maximising a local performance metric — customer acquisition rate, short-term contribution margin, churn reduction — will systematically sacrifice long-run compounding when the measurement horizon of the objective function is shorter than the compounding horizon of the economic architecture. This is not a principal-agent problem in the classical sense; it is an objective function design problem. The relevant question is not whether the agent can be monitored but whether the agent's objective function captures the correct long-run dependencies.

2.3 Dynamic Capabilities: Necessary but Insufficient

Teece, Pisano and Shuen's dynamic capabilities framework addressed the recognised limitation of both transaction cost economics and Porter's positional framework: neither explains how firms sustain advantage under rapid environmental change. Dynamic capabilities — defined as the

[7]Brynjolfsson, E. & McAfee, A. (2014). The Second Machine Age. W.W. Norton, New York.

capacity to sense, seize, and reconfigure resources in response to environmental shifts — provide the adaptive mechanism that static frameworks omit.[8]

The dynamic capabilities framework is a necessary component of any theory of the AI-era firm: firms must be able to reconfigure in response to the rapid obsolescence that AI acceleration creates. However, it does not specify how cross-domain optimisation should be coordinated when each capability domain is locally optimisable by AI. A firm with strong dynamic capabilities in each of its four competitive dimensions — segmentation, contribution architecture, cost physics, and customer lifetime value — can still exhibit poor total economic performance if the dimensions are optimised independently and the binding constraint is not identified or addressed. Dynamic capabilities explain adaptation; they do not explain constraint arbitration.[9]

The insufficiency is structural rather than incidental. Dynamic capabilities theory operates at the level of the individual capability or resource bundle. The multiplicative architecture framework operates at the level of the interaction between dimensions — specifically, the cross-dimensional coupling effects that produce emergent system behaviour that cannot be predicted from the properties of individual dimensions in isolation. A firm that has optimised each dimension independently may exhibit a cross-dimensional coupling that reverses the expected sign of a local improvement, producing aggregate outcomes that individual capability analysis cannot anticipate.

2.4 The Resource-Based View and AI-Accelerated Erosion

Barney's resource-based view proposed that sustained competitive advantage derives from resources that are valuable, rare, inimitable, and non-substitutable. The framework's predictive

[8]Agrawal, A., Gans, J. & Goldfarb, A. (2018). Prediction Machines: The Simple Economics of Artificial Intelligence. Harvard Business Review Press, Boston.
[9]Goldfarb, A. & Tucker, C. (2019). Digital economics. Journal of Economic Literature, 57(1), 3–43.

accuracy rests on the empirical assumption that resource inimitability is durable — that rivals cannot rapidly replicate the resources that confer advantage.[10]

AI accelerates imitation along several dimensions simultaneously. Large language models and generative AI systems enable rapid replication of knowledge-intensive tasks previously considered firm-specific. AI-assisted product development compresses the time from concept to market. Data network effects — a primary source of AI-era inimitability — concentrate advantage in early-scale platforms but provide limited protection for smaller firms whose data advantages are quickly eroded by platform aggregation. The primary domain of durable inimitability in AI-mediated markets is therefore not the individual resource or capability but the architecture of the economic relationship between the firm and its customer base — a system of mutual dependencies that accumulates over time and resists replication because it is built on relational history rather than transferable assets.[11]

3. CUSTOMER ECONOMIC ARCHITECTURE AS A MULTIPLICATIVE SYSTEM

3.1 Formal Specification

Let the competitive architecture of a firm be represented by four normalised domain scores $D_i \in [0, 1]$, where 1 represents best-practice benchmark performance and 0 represents structural failure:

D_{WHO} : economic segment precision and ecosystem positioning

[10]Autor, D., Levy, F. & Murnane, R.J. (2003). The skill content of recent technological change: An empirical exploration. Quarterly Journal of Economics, 118(4), 1279–1333.
[11]Acemoglu, D. & Restrepo, P. (2022). Tasks, automation, and the rise in US wage inequality. Econometrica, 90(5), 1973–2016.

D_WHAT : contribution margin architecture and offer defensibility

D_HOW : cost physics and sub-linear operational leverage

D_CLV : customer lifetime value compounding and expansion depth

Total economic value per customer cohort over a defined time horizon is approximated as:

$$TEV \approx \prod_i D_i = D_WHO \times D_WHAT \times D_HOW \times D_CLV \qquad (1)$$

where TEV denotes the long-run compounded economic output of the customer architecture — the expected present value of the multi-period contribution margin stream generated by the customer base, net of acquisition and service costs. For operational use, each D_i is scored on a 0–10 integer scale; expression (1) applies to the normalised [0,1] values.

3.2 Three Non-Trivial Properties

The multiplicative specification generates three predictions that additive alternatives do not. First, weakest-link dominance: the partial derivative of TEV with respect to any dimension D_i is:

$$\partial TEV / \partial D_i = \prod_{j \neq i} D_j \qquad (2)$$

This implies that the marginal value of improving any dimension equals the product of all other dimension scores. When one dimension approaches zero, its marginal improvement value also approaches zero — but the marginal improvement value of the near-zero dimension itself exceeds that of any other dimension. The binding constraint has the highest marginal return to improvement, provided the other dimensions are held constant. This is the formal expression of Goldratt's constraint dominance applied to economic architecture.

Second, balanced uplift dominates concentrated uplift. Consider a resource budget sufficient to improve total dimension scores by 1.0 units. Under the additive model, the allocation is irrelevant: each unit of improvement adds equally to total output. Under the multiplicative model, the optimal allocation is the one that maximises $\prod_i (D_i + \delta_i)$ subject to $\Sigma\delta_i = 1.0$. By the AM-GM inequality, this product is maximised when the allocation is concentrated on the binding constraint — not when distributed uniformly. The multiplicative model therefore predicts that constraint-first allocation strictly dominates uniform allocation whenever dimension scores are unequal.[12]

Third, non-compensability: no finite strength in one dimension can offset a near-zero score in another, because the product of any positive number with zero is zero. This prediction distinguishes the multiplicative model from lexicographic, additive, and compensatory multi-attribute models, all of which allow strength in one dimension to substitute for weakness in another. The empirical question — whether the multiplicative or additive specification better fits firm-level outcome data — is an open research question identified in Section 10.

3.3 Cross-Domain Coupling

The strict independence assumption embedded in expression (1) is violated in practice. The domains exhibit systematic cross-elasticities that must be modelled before any unilateral domain improvement is implemented:

$$\partial D_CLV/\partial D_HOW < 0 \quad \text{(in high-consideration categories, cost reduction degrades retention)}$$

$$\partial D_WHAT/\partial D_WHO > 0 \quad \text{(sharper segmentation increases willingness-to-pay)}$$

[12]Rochet, J.C. & Tirole, J. (2003). Platform competition in two-sided markets. Journal of the European Economic Association, 1(4), 990–1029.

$$\partial D_CLV/\partial D_WHAT > 0 \quad \text{(richer contribution architecture enables expansion)}$$

$$\partial D_WHO/\partial D_CLV > 0 \quad \text{(deeper relationships amplify referral and ecosystem positioning)}$$

(3)

The first coupling in expression (3) is the most consequential and least intuitive. In high-consideration categories — financial services, healthcare, complex B2B services — customer retention is substantially determined by trust capital: the accumulated perception of reliable performance under complexity. When HOW cost reduction eliminates service capacity that generates trust capital, the D_CLV damage exceeds the D_HOW saving in net present value terms over a twenty-four to thirty-six month horizon. This cross-domain coupling has a negative sign, reversing the expected sign of the local HOW improvement. The time lag between the HOW saving and the CLV cost — the latter appearing in cohort retention curves twelve to twenty-four months later — is the mechanism by which this coupling is systematically underweighted in short-cycle governance.

4. AI CAPABILITY AND THE LOCAL OPTIMISATION PROBLEM

4.1 What AI Can Optimise

Contemporary AI systems — including large language models, reinforcement learning systems, and predictive analytics platforms — exhibit strong capability for local optimisation within defined objective functions. Within each domain of the Box Model, AI enables the following advances: in WHO, micro-segmentation at scale, real-time cluster identification, and propensity-to-buy

prediction with accuracy that manual analysis cannot approach; in WHAT, dynamic pricing, automated bundling optimisation, and personalised value proposition generation; in HOW, autonomous cost allocation, demand-responsive capacity management, and fraud detection that reduces operational loss; in CLV, churn prediction with lead times sufficient for proactive intervention, next-product recommendation at scale, and personalised re-engagement sequencing.[13]

The economic significance of these capabilities is substantial. Eloundou et al. demonstrate that large language model capabilities overlap with a significant fraction of current knowledge-worker task bundles, implying that many firm-specific advantages previously attributable to skilled labour are increasingly replicable through AI augmentation. For each individual domain, AI therefore reduces the advantage of internalisation relative to market provision.[14]

4.2 What AI Cannot Optimise: The Meta-Problem

Despite strong within-domain performance, AI systems face a structural limitation at the level of domain interaction. This limitation is not a current capability deficit that will be resolved by future model generations; it is a structural feature of objective-function-based optimisation systems.

An AI system maximising D_HOW — minimising cost-to-serve — will, under standard gradient descent or reinforcement learning, systematically reduce service capacity wherever the within-domain objective function permits. It will not, absent explicit cross-domain constraint specification, model the second-order D_CLV consequences of that reduction. The cross-domain coupling in expression (3) — specifically the negative $\partial D_CLV/\partial D_HOW$ term — is not a

[13]Parker, G.G., Van Alstyne, M.W. & Choudary, S.P. (2016). Platform Revolution. W.W. Norton, New York.
[14]Goldratt, E.M. & Cox, J. (1984). The Goal: A Process of Ongoing Improvement. North River Press, Great Barrington.

variable within the AI's defined optimisation problem unless it has been explicitly included by the governance layer.

This is the meta-problem: AI optimises within defined objective functions with precision and speed that human management cannot match. But the selection of which objective function is appropriate at the system level — which domain constrains total value, which cross-domain couplings must be modelled before any local optimisation is executed, and which short-term local efficiency should be sacrificed to protect long-term compounding — is a governance problem that requires information about the full multiplicative architecture, not just the local domain.

The meta-problem has a second dimension: commitment. Constraint governance sometimes requires accepting visible short-term deterioration in a non-binding domain to protect the binding constraint. An AI system instructed to maximise total economic value without cross-domain coupling information will typically sacrifice long-run CLV compounding to protect short-run HOW efficiency, because the HOW outcome is within-period observable while the CLV consequence is multi-period lagged. Commitment to accepting short-term local inefficiency for long-term architectural integrity is an institutional property, not a computational one.

4.3 Stylised Empirical Illustrations

Retail financial services: The service-capacity trap

A large retail bank implements AI-driven customer service automation, achieving a thirty-five percent reduction in cost-to-serve (D_HOW improvement of +0.15 on the normalised scale). The AI system correctly identifies low-utilisation relationship manager capacity as the primary cost driver. Over the following eighteen months, cohort-level analysis reveals a twelve percent increase in annual churn among the top quintile of customers by lifetime value — the cohort most

sensitive to perceived service quality deterioration. The NPV of the D_CLV loss exceeds the D_HOW saving by a factor of 2.3 over a five-year horizon at an eight percent discount rate. The AI optimised correctly within its defined objective function. The governance failure was the absence of cross-domain coupling modelling before implementation.

E-commerce: Algorithmic segmentation and margin erosion

A digital retail platform deploys AI micro-segmentation to maximise D_WHO precision, achieving a forty percent improvement in acquisition targeting efficiency (reduced cost per acquired customer across defined persona clusters). The AI correctly identifies high-conversion segments. However, the targeted segments exhibit lower baseline contribution margins — the high-conversion customers are price-sensitive deal-seekers rather than high-lifetime-value regulars. The D_WHO improvement is real. The net TEV effect is negative because the improvement is in the wrong segment type: volume-maximising rather than amplification-targeting. The AI optimised the metric it was given. The metric was wrong.

SaaS: Cross-domain compounding when architecture is coherent

A mid-market SaaS platform maintains coherent cross-domain architecture through explicit constraint governance: quarterly scoring of all four domains, constraint-first capital allocation, and explicit cross-domain impact modelling before any product or pricing decision. Over four years, the platform achieves net revenue retention of one hundred fourteen percent — a D_CLV configuration that generates compounding revenue growth without new-customer acquisition investment. The multiplicative architecture compounds: strong D_CLV reduces effective D_HOW cost (retained customers require less reacquisition spend), which creates contribution margin oxygen that funds D_WHAT expansion, which reinforces D_CLV through

switching cost accumulation. AI augments each domain; governance maintains the multiplicative integrity.[15]

5. FIVE PROPOSITIONS: THE AI-ERA THEORY OF THE FIRM

We now formalise the foregoing analysis into five propositions characterising the AI-era firm. Each proposition follows from the combination of the AI capability analysis in Section 4 and the multiplicative architecture framework in Section 3.

Proposition 1: The Friction Erosion Boundary Condition

As AI reduces the transaction costs that justify firm internalisation in a domain, the explanatory power of coordination-based firm theory for that domain declines. The remaining explanatory domain of classical firm theory contracts to activities where asset specificity, relational capital, and temporal commitment generate returns that AI-augmented markets cannot replicate. The boundary of the firm should contract toward these residual advantages.

Proposition 1 is consistent with the historical trajectory of outsourcing and platform disintermediation, but advances a more precise prediction: the rate of boundary contraction should track the rate of AI capability expansion in each task domain. The implication is that firm boundary decisions must be revisited continuously rather than periodically, and that domains in which AI has achieved strong local optimisation capability should be candidates for market provision unless cross-domain coupling arguments justify retention.

[15]Reichheld, F.F. & Sasser, W.E. (1990). Zero defections: Quality comes to services. Harvard Business Review, 68(5), 105–111.

Proposition 2: The Architecture Shift in Competitive Advantage

In AI-mediated markets, competitive advantage shifts from friction management to compounding architecture governance. Advantage accrues not to the firm that best reduces the transaction costs of exchange, but to the firm that best maintains the multiplicative integrity of its customer economic architecture across time and competitive perturbation.

Proposition 2 extends D'Aveni's transient advantage observation: the relevant axis of advantage is not the duration of positional advantage (which contracts under AI) but the quality of the architectural compounding that the firm generates over time. A firm with lower per-period positional advantage but superior architectural compounding — specifically, a D_CLV configuration that generates expansion revenue exceeding churn losses, combined with sub-linear HOW cost physics — can outperform a firm with higher positional advantage but poor architectural integrity over a multi-period horizon.[16]

Proposition 3: Local Optimisation and Architectural Fragility

In multiplicative economic systems, AI-enabled local optimisation of individual domains increases systemic fragility when cross-domain constraint arbitration is absent. The mechanism is the negative cross-domain coupling identified in expression (3): unilateral domain optimisation that ignores second-order effects on adjacent domains can reduce total economic value even when it improves the targeted domain score.

Proposition 3 generates a specific and testable prediction: firms that deploy AI optimisation tools at the domain level without cross-domain governance protocols will exhibit higher variance in D_i scores over time — increasing imbalance scores — and correspondingly higher margin volatility, because the multiplicative architecture becomes progressively more fragile as the

[16]Gupta, S. & Lehmann, D.R. (2005). Managing Customers as Investments. Wharton School Publishing, Upper Saddle River.

binding constraint tightens while non-binding domains are optimised. This prediction is directly testable against firm-level financial data.

Proposition 4: The Primary Function of Hierarchy in the AI Era

The primary function of organisational hierarchy in AI-mediated markets is constraint declaration and capital reallocation across domains — the governance functions that AI optimisation systems cannot perform because they require information about the full multiplicative architecture, cross-domain coupling effects, and institutional commitment to accepting short-term local inefficiency for long-term compounding integrity.

Proposition 4 reframes the Coasian and Simonian accounts of hierarchy. Where Coase explained hierarchy as a coordination substitute for costly markets, and Simon identified hierarchy as a satisficing mechanism for bounded rationality, we propose a third account: hierarchy as a commitment device for multi-period architectural governance in the presence of AI systems that are locally superior at optimisation but structurally blind to cross-domain coupling effects and their long-run compounding consequences.[17]

Proposition 5: Dissolution Risk Under Absent Architectural Governance

Firms that fail to institutionalise cross-domain architectural governance will, under AI-mediated competitive conditions, tend toward dissolution into modular market networks. Without constraint arbitration, each domain will be locally optimised — eventually by AI systems at near-zero cost — eliminating the relative advantage of internalisation without replacing it with the architectural compounding that requires governance. The firm dissolves into a network of AI-optimised market transactions.

[17]Katz, M.L. & Shapiro, C. (1985). Network externalities, competition, and compatibility. American Economic Review, 75(3), 424–440.

Proposition 5 articulates the boundary condition of the theory: the AI-era firm is not generically advantaged over AI-augmented markets. It is advantaged specifically in the constraint governance and compounding architecture functions identified in Proposition 4. Firms that cannot or do not perform these functions will find their internalisation advantages progressively eroded by AI, and their optimal organisational form will shift toward platform participation or market provision rather than hierarchical coordination.

6. CONSTRAINT GOVERNANCE AS THE CORE ORGANISATIONAL FUNCTION

6.1 The Constraint Governance Problem

Constraint governance is the organisational process of (i) identifying the binding domain in the multiplicative architecture at a given point in time, (ii) modelling the cross-domain consequences of proposed interventions before implementation, (iii) reallocating capital asymmetrically toward the binding constraint, and (iv) committing to accepting temporary deterioration in non-binding domains as the price of constraint removal.

This process differs from classical strategic planning in three respects. First, it is constraint-first rather than balanced: the deliberate intent is to identify the single binding domain and concentrate resources on it, not to improve all domains simultaneously. Second, it is cross-domain by design: no domain intervention can be approved without prior modelling of its second-order effects on adjacent domains. Third, it requires institutional commitment to temporal misalignment: the short-term cost of constraint removal appears in the period of intervention, while the

compounding benefit appears in subsequent periods with a lag that can exceed twelve to twenty-four months.

The third requirement — institutional commitment to temporal misalignment — is the most distinctive and the most analytically significant. It is not a computational problem. AI systems can be instructed to model multi-period outcomes and to weight long-term compounding over short-term efficiency. The commitment problem arises from the governance structure within which resource allocation decisions are made: quarterly reporting cycles, domain-indexed performance metrics, and principal-agent incentive structures that reward near-term attribution all operate against constraint governance. The constraint governance function therefore requires not merely analytical capability but institutional architecture — governance design that creates the conditions under which constraint-first allocation is incentive-compatible for managers.[18]

6.2 The Activity Trap as Governance Pathology

The Activity Trap is the systematic organisational preference for visible, attributable, short-cycle domain improvements over cross-domain constraint removal. It is a predictable outcome of the principal-agent incentive structure identified in Section 2.2: managers rewarded for near-term attribution will systematically prefer domain-level activity — launching campaigns, deploying tools, reducing headcount — over constraint identification and architectural recalibration, which reduces short-term predictability and extends attribution cycles.

The Activity Trap produces a characteristic signature in the multiplicative architecture: improvement in non-binding domain scores combined with deterioration or stagnation in the binding constraint domain. Because the multiplicative relationship concentrates the marginal

[18]Simon, H.A. (1991). Organizations and markets. Journal of Economic Perspectives, 5(2), 25–44.

return on the binding constraint, this pattern produces flat or declining TEV despite increasing activity levels in non-binding domains. Firms trapped in this dynamic appear operationally active while their architectural compounding capacity declines.

The governance design implication is direct: the Activity Trap cannot be resolved by adding analytical capability to the governance system. It requires changing the incentive structure under which resource allocation decisions are made — specifically, replacing domain-indexed performance metrics with architecture-level metrics (TEV, imbalance score, NRR, constraint identification accuracy) and making cross-domain impact modelling a governance requirement rather than an optional analytical input.

6.3 The External Moderation Requirement

A structural feature of constraint governance in organisations is the impossibility of internal constraint identification under standard incentive conditions. The domain whose resource allocation should be reduced — because it is non-binding and its improvement does not increase TEV — is typically managed by individuals whose career interests align with protecting that domain's resources. Internal constraint identification processes therefore produce systematically defensive results: the reported binding constraint is rarely the domain managed by the most powerful internal stakeholder.

External moderation — the introduction of a party with no stake in domain resource allocation and with full access to cohort-level economic data — is therefore a structural requirement of constraint governance, not a procedural preference. This is a direct implication of the agency cost analysis: the Information about true domain scores is asymmetrically distributed in

favour of domain managers, creating an adverse selection problem that internal governance processes cannot resolve without structural intervention.

7. BOUNDARY CONDITIONS AND SCOPE

A theory of the firm is a scientific claim whose scope must be specified. The compounding architecture framework applies under conditions that can be stated with reasonable precision, and its non-application can similarly be characterised.

7.1 Where the Theory Applies

The compounding architecture framework generates its strongest predictions in markets exhibiting the following features: first, recurring or relationship-based economic exchange, in which customer retention and expansion generate multi-period compounding dynamics that are materially larger than single-period transaction value; second, significant customer economic heterogeneity, in which the distribution of contribution margin, cost-to-serve, and lifetime value across the customer base is sufficiently concentrated that top-decile customers generate a disproportionate share of firm economic value; third, partial AI tractability, in which AI can optimise individual competitive domains but cannot itself perform the cross-domain constraint arbitration function; and fourth, governance-sensitive compounding, in which the long-run compounding rate is materially affected by governance decisions — specifically, by the quality of constraint identification and capital reallocation.

These conditions are characteristic of, but not exclusive to, digital subscription businesses, retail financial services, enterprise software, healthcare services, and other high-consideration relationship-based categories. They are less characteristic of commodity spot markets, regulatory monopolies, and single-transaction exchange platforms.

7.2 Where Classical Theory Retains Priority

Classical firm theory retains full predictive accuracy in three structural configurations. First, in markets with high physical asset specificity and capital intensity — regulated utilities, extractive industries, capital-intensive manufacturing — the transaction cost and governance logic of Williamson retains its explanatory power because AI does not reduce physical asset specificity. Second, in procurement-dominated B2B markets with institutionally determined switching costs — public sector contracting, regulated financial intermediaries, defence procurement — the Five Forces framework accurately predicts margin structures because industry structure, not customer architecture, is the primary determinant of profit. Third, in markets with legally mandated interoperability and minimum service standards, where the compounding advantage of relationship depth is structurally capped.[19]

The theory is also inapplicable in pre-revenue or early-stage ventures where customer economic architecture does not yet exist in sufficient depth to generate compounding dynamics. In these contexts, classical theories of market entry, competitive positioning, and resource acquisition are the appropriate analytical instruments.

[19]Eisenhardt, K.M. & Martin, J.A. (2000). Dynamic capabilities: What are they? Strategic Management Journal, 21(10–11), 1105–1121.

7.3 The Governance Layer vs. the AI Layer

A potential misreading of the theory is that it positions AI and governance as alternatives — that firms must choose between AI optimisation and architectural governance. The correct relationship is complementary but hierarchical: AI operates as a domain-level optimisation layer that executes within objective functions; governance operates as the meta-layer that sets objective functions, identifies constraints, and manages cross-domain coupling.

The governance layer does not replace AI; it directs it. Strong constraint governance improves the return on AI investment by ensuring that AI optimisation is concentrated in the binding constraint domain and that cross-domain coupling effects are explicitly modelled before AI-driven domain changes are implemented. Weak governance does not reduce AI capability; it misdirects it, concentrating AI optimisation in non-binding domains while the binding constraint continues to suppress TEV.

8. IMPLICATIONS FOR ORGANISATIONAL DESIGN

The five propositions and the constraint governance model generate a set of organisational design implications that differ materially from both the efficiency-focused recommendations of transaction cost economics and the capability-development focus of dynamic capabilities theory.[20]

8.1 Separating the AI Optimisation Layer from the Governance Layer

The most structurally significant design implication is the requirement to separate AI-driven domain optimisation from architectural governance. Organisations that allow AI optimisation

[20]Kaplan, R.S. & Norton, D.P. (1996). The Balanced Scorecard. Harvard Business School Press, Boston.

systems to operate without a governance layer capable of cross-domain constraint modelling will systematically generate the pathology described in Proposition 3: local domain improvement accompanied by architectural fragility.

The practical implementation requires distinct reporting structures, objective functions, and decision rights for each layer. The AI optimisation layer has domain-specific metrics as its objective function — cost-per-acquisition for the HOW domain, churn probability for the CLV domain — and operates within boundaries set by the governance layer. The governance layer has TEV, architectural imbalance score, and constraint identification accuracy as its objective function, and sets the cross-domain coupling constraints within which domain-level AI operates.

8.2 Architecture-Level Metrics as Primary Governance Instruments

Current governance practice — in both corporate boards and management teams — concentrates on domain-level metrics: revenue growth, cost efficiency, customer acquisition rate, margin percentage. These metrics are accurate within-domain, but they do not capture the multiplicative interaction between domains that determines TEV. Kaplan and Norton's balanced scorecard represented an early attempt to aggregate multiple performance dimensions; the constraint architecture framework requires a further step: not merely measuring multiple dimensions, but modelling their multiplicative interaction and identifying the binding constraint that should receive asymmetric resource allocation.[21]

The primary governance metrics for an AI-era firm are therefore: TEV score (the product of normalised domain scores), architectural imbalance score (the standard deviation of domain scores, measuring structural fragility), net revenue retention by cohort (the primary D_CLV

[21]D'Aveni, R.A. (1994). Hypercompetition: Managing the Dynamics of Strategic Maneuvering. Free Press, New York.

metric), cost per profitable lifetime cohort (the primary D_HOW metric), and constraint identification accuracy (the frequency with which the declared binding constraint matches the domain that received resource allocation in the prior period). These metrics are not supplementary to traditional financial reporting; they are leading indicators of the financial outcomes that traditional metrics will report with a twelve-to-twenty-four-month lag.

8.3 Temporal Architecture of Governance Cadences

Constraint governance requires multiple temporal cadences operating in parallel. A quarterly identification protocol updates domain scores using current cohort-level data, declares the binding constraint for the period, models cross-domain effects of proposed interventions, and reallocates discretionary capital toward the identified constraint. An annual deep intervention recalibrates the full economic architecture — re-scoring all domains, revalidating the cross-domain coupling structure, and assessing whether the ecosystem and competitive conditions have shifted sufficiently to alter the architecture's strategic logic. A multi-year strategic horizon — five to ten years — maintains focus on the compounding dynamics that quarterly governance cannot fully capture.

The temporal architecture is not bureaucratic but emergent: organisations that sustain constraint governance over multiple cycles develop the institutional knowledge necessary to identify binding constraints earlier, model cross-domain effects more accurately, and commit to constraint removal more decisively. Constraint governance is therefore a dynamic capability in the sense of Teece: it improves with practice and generates compounding returns at the governance level as well as the economic architecture level.

9. MATHEMATICAL APPENDIX: FORMAL PROPERTIES OF THE MULTIPLICATIVE ARCHITECTURE

9.1 Optimal Allocation Under the Multiplicative Constraint

Let the firm's resource budget be normalised to B = 1.0. Let current domain scores be $D = (D_1, D_2, D_3, D_4)$ and let allocation $\delta = (\delta_1, \delta_2, \delta_3, \delta_4)$ with $\Sigma_i \delta_i = B$. Assume linear domain response functions: $\tilde{D}_i = D_i + \delta_i$. The optimisation problem is:

$$\max_{\{\delta\}} \prod_i (D_i + \delta_i) \text{ subject to } \Sigma_i \delta_i = B, \ \delta_i \geq 0 \qquad (A1)$$

Taking logarithms, this is equivalent to:

$$\max_{\{\delta\}} \Sigma_i \ln(D_i + \delta_i) \text{ subject to } \Sigma_i \delta_i = B \qquad (A2)$$

The Lagrangian is:

$$\mathscr{L} = \Sigma_i \ln(D_i + \delta_i) - \lambda(\Sigma_i \delta_i - B) \qquad (A3)$$

First-order conditions: $1/(D_i + \delta_i) = \lambda$ for all i. This implies $\tilde{D}_i = 1/\lambda$ for all i — all domain scores are equalised at the optimum. The Lagrange multiplier is $\lambda = n/（\Sigma_i D_i + B)$.

The optimal allocation concentrates resources on the lowest-scoring dimension until it equals the second-lowest, then jointly increases both until they equal the third-lowest, and so on — a water-filling algorithm. This formal result confirms the intuition of Proposition 4: constraint-first capital allocation, implemented iteratively until all domain scores are equalised, maximises TEV under the multiplicative specification.

9.2 The Value of Constraint Identification

Let $D_min = \min(D_1, D_2, D_3, D_4)$ be the binding constraint domain score and $D_\{-min\} = \prod_{j \neq min} D_j$ be the product of the non-binding domain scores. $TEV = D_min \times D_\{-min\}$.

The value of correctly identifying the binding constraint, relative to uniform allocation, is:

$$VCBI = TEV^*(B|\text{correct ID}) - TEV^*(B|\text{uniform}) = [D_min + B] \times D_\{-min\} - [\prod_i (D_i + B/n)] \quad \text{(A4)}$$

where $TEV^*(B|\text{correct ID})$ is the TEV achieved by concentrating the full budget B on the binding constraint, and $TEV^*(B|\text{uniform})$ is TEV under equal allocation across all four domains. When $D_min << D_\{-min\}/n$, the value of constraint identification VCBI is large. When domain scores are approximately equal (balanced architecture), VCBI approaches zero and uniform allocation is near-optimal. This formalises the prediction that constraint governance is most valuable in architecturally imbalanced firms — precisely the firms that are most prevalent in practice.

9.3 Dynamic Extension: Compounding Under CLV

Let the CLV domain generate intertemporal compounding through the customer lifetime value mechanism. Define the multi-period TEV as:

$$TEV_T = D_WHO \times D_WHAT \times D_HOW \times [\Sigma_{t=0}^{T} (AOV_t \times f_t \times m_t \times \rho^t)] \quad \text{(A5)}$$

where ρ = annual retention rate, T = planning horizon, and the CLV term is the discounted sum of per-period customer contribution. The partial derivative of TEV_T with respect to ρ is:

$$\partial TEV_T/\partial \rho = D_WHO \times D_WHAT \times D_HOW \times [\Sigma_{t=0}^{T} t \times AOV_t \times f_t \times m_t \times \rho^{t-1}] \quad \text{(A6)}$$

Expression (A6) is increasing in t — the marginal value of retention improvement increases with the planning horizon. This formalises the result that the CLV domain has superlinear leverage over TEV_T: a given improvement in retention rate ρ produces larger TEV gains as the planning horizon extends, because the compounding effect accumulates across periods. This is the formal basis for the empirical observation of Reichheld and Sasser that retention improvements produce disproportionate long-run profit impact relative to other margin improvements.

9.4 Parametric Assumptions and Sensitivity

All numerical illustrations in the paper use the parametric baseline in Table 1. Table 2 presents sensitivity of five-year compounded TEV to retention improvement across baseline conditions.

Parameter	Baseline Value
Annual gross retention rate	75–85%
Gross contribution margin	40–60%
Simulation horizon	5–10 years
Discount rate	8–12%
Expansion penetration (annual)	15–25% of existing base
D_i normalisation	[0,1]; operational scoring 0–10 integer scale

Table 1. Parametric baseline for all numerical illustrations.

Baseline Retention / Margin	+2pp	+5pp	+8pp	+10pp
70% retention / 40% margin	+11%	+26%	+44%	+58%
80% retention / 50% margin	+14%	+31%	+52%	+68%
85% retention / 60% margin	+18%	+40%	+66%	+87%

Table 2. Sensitivity of 5-year compounded TEV to retention improvement (pp = percentage points). Assumes D_WHO, D_WHAT, D_HOW held constant.

10. CONCLUSION AND RESEARCH AGENDA

10.1 Summary

This paper has advanced a theory of the firm grounded not in the minimisation of market friction but in the governance of compounding economic architecture under constraint. The argument proceeds in three steps. First, AI systematically erodes the frictions on which classical firm theory rests — reducing search costs, compressing information asymmetry, automating coordination, and enabling domain-level local optimisation at near-zero cost. This erosion narrows the explanatory domain of transaction cost economics, agency theory, and structural positioning frameworks without eliminating it.

Second, the multiplicative architecture framework formalises firm value as a product of four interdependent domain scores, generating non-trivial predictions: weakest-link constraint dominance, diminishing returns to unilateral domain optimisation, and non-compensability across dimensions. These predictions distinguish the multiplicative model from additive alternatives and provide the theoretical foundation for constraint-first capital allocation as the governing logic of strategic resource deployment.

Third, the five propositions characterise the AI-era firm as a constraint-orchestrating governance structure that performs the meta-optimisation function that AI domain-level systems cannot execute: cross-domain constraint identification, coupling effect modelling, and institutional

commitment to accepting temporal misalignment between short-term cost and long-term compounding benefit.

The theory preserves classical contributions by identifying their boundary conditions rather than refuting them. Transaction cost economics retains full validity in domains where AI has not reduced the four Coasian cost categories to near-zero. Porter's positional framework retains predictive accuracy in structurally bounded, asset-intensive industries. Dynamic capabilities theory is incorporated as the adaptive mechanism that enables architectural recalibration under environmental change. The novel contribution is the multiplicative synthesis and its governance translation.

10.2 Research Agenda

The theory generates five empirically testable research directions:

(1) Multiplicative versus additive predictive accuracy. Cross-sectional and longitudinal empirical tests comparing the out-of-sample predictive accuracy of multiplicative and additive specifications for firm-level profitability trajectories would provide the foundational validation evidence for the central specification claim. The test requires firm-level scoring of the four domains and subsequent comparison of predicted versus actual financial outcomes across both specifications.

(2) Cross-domain elasticity estimation. Econometric estimation of the coupling coefficients in expression (3) — particularly the D_HOW-to-D_CLV negative coupling — across subscription and transactional sectors would both validate the theoretical structure and provide calibration parameters for the constraint governance simulation tools implied by the framework. Panel data on

service capacity decisions and subsequent cohort-level churn rates provides the natural experimental context.

(3) Architectural imbalance as a predictor of margin volatility. If the standard deviation of normalised domain scores measures structural fragility, architecturally imbalanced firms should exhibit higher margin volatility and earnings revision frequency. This prediction is testable against firm-level financial data using analyst forecast revisions as a proxy for governance-induced earnings uncertainty.

(4) AI deployment and architectural fragility. The theory predicts that AI domain-level deployment without governance protocols increases architectural imbalance scores over time, as non-binding domains are optimised while binding constraints are not addressed. Natural experiments in AI adoption — using the staggered rollout of AI optimisation tools across firms and divisions — provide identification for this prediction.

(5) Governance intervention and Activity Trap behaviour. Experimental and quasi-experimental analysis of governance mechanisms — external facilitation requirements, constraint-indexed incentive structures, cohort-level transparency mandates — that reduce Activity Trap behaviour would validate the agency cost mechanism and identify effective institutional designs. The constraint identification accuracy metric proposed in Section 8.2 provides a continuous outcome variable for such studies.

10.3 Closing Argument

The classical firm persisted because markets were imperfect. The AI-era firm persists because architectures are complex.

AI accelerates; firms commit. AI optimises domains; firms arbitrate constraints. AI maximises objective functions; firms choose which objective functions to set. The locus of irreplaceable organisational value moves from coordination — which AI can substitute — to constraint governance — which AI cannot perform without a prior governance layer that specifies the full multiplicative architecture and accepts the temporal misalignment between short-term cost and long-term compounding benefit.

The theoretical implication is that the future of the firm is not its dissolution into AI-augmented market networks, but its transformation into a more focused governance structure: smaller in operational scope as AI substitutes for coordination tasks, but more consequential in the architectural decisions it makes and defends over time. The firms that thrive in an AI economy will not be those that deploy AI most aggressively in every domain. They will be those that govern the multiplicative architecture that AI operates within — identifying constraints before AI optimises around them, modelling cross-domain consequences before AI executes local improvements, and committing to long-term compounding when AI recommends short-term efficiency.

AI optimises. Firms choose. The quality of the choice determines the quality of the compound.

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